



prevalent weather conditions at the time of farming, and water sources.

This data is made available to farmers easily through a mobile application. Farmers can verify this 'analysed' data with government authorities. This eliminates the need of middlemen as exact prices are computed smartly.

For example, "GreenAgtech provides a platform to sell the harvest to HORECA (hotels, restaurants, cafe segment), who can place order via the app. After the purchase order is placed, the company supplies fruits and vegetables to the order placer. The best part about such a platform is that farmers get the best prices without time delays," shares Shweta Gaded, business analyst at Bengaluru-based Inventrom (a company that develops and deploys Bolt agro-IoT solution).

According to a seasoned technology professional (who offered insights on the condition of anonymity), apart from the agriculturist community, IoT researchers get to benefit the most as they get the opportunity to develop easily implementable IoT-driven agricultural solutions.

"Hire engineers by the lot to work for your agro-IoT startups. Implement sensors at the soil level. These sensors will send data that helps in determining a farmer's future. Vital aspects such as soil quality, moisture level and temperature can be analysed through Big Data and analytics, and delivered (through text messages) to farmers as customised dashboards," he advises agro-IoT startups.

Making existing markets smart

Though at a nascent stage, agro-IoT solutions are completely practical, feel embedded engineers.

IoT devices are not uncommon in marketplaces. These have seen the light of the day in several western economies. However, there are hitches. Geographical reach and connectivity with rural areas is still in its infancy in India. Marketplaces and fair-price shops are generally located

Major experts consulted

- Ravi Gupta, business analyst at Inventrom
- Dr Regi Mathew, chief data scientist at Clever Insight
- Swetha Gaded, business analyst at Bengaluru-based Inventrom

50-100 kilometres away from the farms where produce is harvested. So farmers still prefer middlemen to sell produce.

Nonetheless, Pune-based startup Ecozen has developed an IoT-driven solar-powered cold storage unit that is accessible even to small and marginal farmers. Tamil Nadu-based Nimble Wireless also offers IoT-driven cold chain solutions to both the food as well as pharma sectors.

"IoT machines allow users to alert their farm for a refill at the push of a button. So middlemen can definitely be eliminated. The onus is now on IoT service providers and embedded engineers to design and deploy such smart market devices," adds Shweta.

Direct open source tools

There are open source platforms that directly connect farmers with traders. So, if you are an engineer intending to start farming on your ancestral land or simply a farmer, check out following open source platforms:

e-NAM. An online portal developed by the government of India for trading in agriculture produce, it promises to liberate farmers from the clutches of middlemen and realise fair market value for their yields.

ConnectFarmer. A common platform for processed farm products, it allows farmers to sell their produce directly to end users.

mKisan. Yet another buyer-seller platform created by the government of India, mKisan enables farmers to connect directly with buyers. Buyers register on this platform and choose the area from where they intend to buy the product. They can quote the price at which they want to procure a product. Farmers of the selected location get a text message on

their mobile handset. This message contains name and mobile number of all interested buyers and the price offered to buy a commodity.

Government initiatives

The government's interest in IoT and smart concepts has seemingly doubled. Together with Nasscom, it has formed the Centre of Excellence on the Internet of Things. This centre aims to further research on IoT across a plethora of domains such as smart cities and smart villages.

"The government is certainly taking necessary steps towards advancing the agriculture technologies and thereby making the life of farmers easier," affirms Ravi Gupta, business analyst at Inventrom.

Other agro-IoT solutions by private players

Agro-IoT solutions have already been deployed at the ground level. A classic example here is the Inventrom Bolt, which is designed to effectively solve the farmers' crop failure issues. It helps farmers with site-specific nutrient management with the help of sensors and environment for crop growth involving storage mapping, and soil quality checking.

Clients use Inventrom's Bolt for a wide variety of applications including farmer profiling, crop monitoring and soil weather monitoring.

"Bolt can save losses due to crop failure. An efficient online marketplace and supply chain management is an optimal solution to enhance farmers' produce sales," signs off Swetha.

For more information on Bolt and its use-cases, readers may please visit the following link: <http://electronicsforu.com/technology-trends/must-read/iot-smart-farming-rescue-farmers> **EFY**